

Cristian Mesquida

POSTDOCTORAL RESEARCHER

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Employment

Postdoctoral researcher

Eindhoven, The Netherlands

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2025-Present

- Contributed to the development and validation of Metacheck, an R package for improving research quality and extracting data from published articles.

PhD candidate

Eindhoven, The Netherlands

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2021-2025

- My PhD thesis examined the factors hindering replicability and reproducibility in sports and exercise science.

Education

PhD

Eindhoven, The Netherlands

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2021-2025

- My PhD thesis focused on the factors that hinder replicability and reproducibility of research in sports and exercise science.

MSc. Integrative Physiology

Barcelona, Spain

UNIVERSITY OF BARCELONA

2017-2019

- My master thesis focused on the effect of two training protocols on maximal oxygen uptake in highly-trained cyclists.

MSc. Research in Sport and Physical Activity

Granada, Spain

UNIVERSITY OF GRANADA

2015-2017

- My master thesis focused on the performance determinants of swimming sprint performance.

BSc. Sports and Exercise Science

Vic, Spain

UNIVERSITY OF VIC

2009-2013

- My bachelor end project focused on developing a training programme to reduce the injury risk of hamstrings in football players.

Publications

- Lakens, D., Mesquida, C., Xavier-Quintais, G., Rasti, S., Toffalini, E., & Altoè, G. (2026). Rethinking Type S and Type M Errors. *Advances in Methods and Practices in Psychological Science*, 9(2), 25152459261432530. <https://doi.org/10.1177/25152459261432530>
- Ditroilo, M., Mesquida, C., Abt, G., & Lakens, D. (2025). Exploratory research in sport and exercise science: Perceptions, challenges, and recommendations. *Journal of Sports Sciences*, 1–13. <https://doi.org/10.1080/02640414.2025.2486871>
- Murphy, J., Caldwell, A. R., Mesquida, C., Ladell, A. J. M., Encarnación-Martínez, A., Tual, A., Denys, A., Cameron, B., Van Hooren, B., Parr, B., DeLucia, B., Mason, B. R. J., Clark, B., Egan, B., Brown, C., Ade, C., Sforza, C., Taber, C. B., Kirk, C., ... Warne, J. P. (2025). Estimating the Replicability of Sports and Exercise Science Research. *Sports Medicine*. <https://doi.org/10.1007/s40279-025-02201-w>
- Lakens, D., Mesquida, C., Rasti, S., & Ditroilo, M. (2024). The benefits of preregistration and Registered Reports. *Evidence-Based Toxicology*, 2(1), 2376046. <https://doi.org/10.1080/2833373X.2024.2376046>
- Mesquida, C., & Lakens, D. (2024). Is the effect large enough to matter? Why exercise physiologists should interpret effect sizes meaningfully: A reply to Williams et al. (2023). *The Journal of Physiology*, 602(1), 241–242. <https://doi.org/10.1113/JP285901>
- Holgado, D., Mesquida, C., & Román-Caballero, R. (2023). Assessing the Evidential Value of Mental Fatigue and Exercise Research. *Sports Medicine*, 53(12), 2293–2307. <https://doi.org/10.1007/s40279-023-01926-w>
- Mesquida, C., Murphy, J., Lakens, D., & Warne, J. (2023). Publication bias, statistical power and reporting practices in the Journal of Sports Sciences: Potential barriers to replicability. *Journal of Sports Sciences*, 41(16), 1507–1517. <https://doi.org/10.1080/02640414.2023.2269357>

8. Murphy, J., Mesquida, C., & Warne, J. (2023). A Survey on the Attitudes Towards and Perception of Reproducibility and Replicability in Sports and Exercise Science. *Communications in Kinesiology*, 1(5). <https://doi.org/10.51224/cik.2023.53>
9. Bossi, A. H., Mesquida, C., Hopker, J., & Rønnestad, B. R. (2023). Adding Intermittent Vibration to Varied-intensity Work Intervals: No Extra Benefit. *International Journal of Sports Medicine*, 44(2), 126–132. <https://doi.org/10.1055/a-1812-7600>
10. Mesquida, C., Murphy, J., Lakens, D., & Warne, J. (2022). Replication concerns in sports and exercise science: A narrative review of selected methodological issues in the field. *Royal Society Open Science*, 9(12), 220946. <https://doi.org/10.1098/rsos.220946>
11. Murphy, J., Mesquida, C., Caldwell, A. R., Earp, B. D., & Warne, J. P. (2022). Proposal of a Selection Protocol for Replication of Studies in Sports and Exercise Science. *Sports Medicine*. <https://doi.org/10.1007/s40279-022-01749-1>
12. Etxebarria, N., Wright, J., Jeacocke, H., Mesquida, C., & Pyne, D. B. (2021). Running Your Best Triathlon Race. *International Journal of Sports Physiology and Performance*, 16(5), 744–747. <https://doi.org/10.1123/ijssp.2020-0838>
13. Bossi, A. H., Mesquida, C., Passfield, L., Rønnestad, B. R., & Hopker, J. G. (2020). Optimizing Interval Training Through Power-Output Variation Within the Work Intervals. *International Journal of Sports Physiology and Performance*, 15(7), 982–989. <https://doi.org/10.1123/ijssp.2019-0260>

Preprints

1. Mesquida, C., Warne, J., & Lakens, D. (2025). *A scoping review of the transparency, reporting practices and methodological rigor of meta-analyses published in Sports Medicine*. SportRxiv. <https://doi.org/10.51224/SRXIV.576>
2. Mesquida, C., Murphy, J., Warne, J., & Lakens, D. (2025). *Prevalence, reporting practices, and methodological quality of a priori power analyses in sports and exercise science research*. SportRxiv. <https://doi.org/10.51224/SRXIV.575>
3. Mesquida, C., Warne, J., & Lakens, D. (2025). *What is your hypothesis? On the importance of knowing your hypothesis before conducting a hypothesis test*. SportRxiv. <https://doi.org/10.51224/SRXIV.577>
4. Mesquida, C., Murphy, J., Warne, J., & Lakens, D. (2025). *On the replicability of sports and exercise science research: Assessing the prevalence of publication bias and studies with underpowered designs by a z-curve analysis*. SportRxiv. <https://doi.org/10.51224/SRXIV.534>

Teaching Experience

Lecturer of Behavioral Research Methods 3

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2025-Present

- Taught p-values and Bayes factors, power analysis, and equivalence testing.

Lecturer of Behavioral Research Methods 2

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2025-Present

- Taught multiple regression and power analysis.

Lecturer of Metascience

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2024-Present

- Taught introduction to meta-analysis

Teaching assistant of Behavioral Research Methods 3

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2024

- Assisted students during tutorial sessions with course assignments

Teaching assistant of Advanced Data Analysis

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2024

- Taught R Markdown for reproducible reporting and data simulation for statistical analysis

Teaching Assistant of Applied Exercise Physiology

TECHNOLOGICAL UNIVERSITY OF DUBLIN

2021

- Taught anatomy

Supervision

Supervisor of Master End Project

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2025

- Co-supervised a project using automated search extraction to extract and attempted hypotheses using large language models

Supervisor of Bachelor End Project

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2025

- Co-supervised a meta-analysis project examining the impact of school sanitation on students' academic performance

Supervisor of Master End Project

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2024

- Co-supervised the development of a tool to identify incorrect use of the standard error of measurement instead of the standard deviation in effect size calculations for meta-analyses

Supervisor of Bachelor End Project

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2023

- Co-supervised four students creating a tool to extract sample sizes from published studies

Research experience

Research internship (unpaid)

Norway

NORWEGIAN SCHOOL OF SPORTS SCIENCE

2018

- Investigated the use of the 1080 Sprint resistance device in swimming to measure power-velocity profiles and estimate drag characteristics in swimmers.

Research internship (unpaid)

Norway

INLAND NORWAY UNIVERSITY OF APPLIED SCIENCES

2017

- Investigated the effects of two HIIT protocols on VO_2max in highly trained cyclists using physiological assessments, including lactate threshold, VO_2max , and cycling economy testing.

Research internship (unpaid)

Spain

UNIVERSITY OF GRANADA

2015-2016

- Contributed to race analysis and the assessment of swimmers' dry-land performance profiles and propulsive force.

Talks

ONH Metascience Symposium

OSLO NYE HØYSKOLE

2026

- Presented findings on the quality of power analyses for sample size justification in sports and exercise science.

Transparency, Technology and AI in Peer Review

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2026

- Presented Metacheck, an automated tool for assessing research transparency.

Metascience Series

CHARLES STURT UNIVERSITY

2025

- Gave a talk on research waste in sports and exercise science.

Workshop on Metacheck

OPEN RESEARCH SUMMER SCHOOL AT KING'S COLLEGE LONDON

2025

- Delivered a workshop on Metacheck, including a presentation and hands-on session.

Danish Reproducibility Network meetup

DANISH REPRODUCIBILITY NETWORK

2025

- Gave a talk on the benefits of preregistration and Registered Reports in hypothesis-testing research.

Lunch meetup for the Open Science Community

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2025

- Gave a talk on the benefits of preregistration and Registered Reports in hypothesis-testing research.

NLRN Pre-Symposium for Young Meta-Scientists

UNIVERSITY MEDICAL CENTER GRONINGEN

2024

- Presented findings on publication bias and statistical power in sports and exercise science.

Paul Meehl Graduate School PhD Day Conference

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2024

- Presented findings on the reporting and reproducibility of a priori power analyses in sports and exercise science.

Workshop on endurance performance

SETANTA COLLEGE

2021-2023

- Conducted a workshop on the physiological determinants of endurance and team-sport performance, including lactate-threshold testing.

Workshop on 1080 Sprint resistance device

KUMMING HIAGENG PHYSICAL TRAINING BASE

2019

- Conducted a workshop on using the 1080 Sprint to assess swimmers' power-velocity profiles and drag

Grants

Replication studies

DUTCH RESEARCH COUNCIL

2026

- Assess the reproducibility of power analyses in large-scale replication projects. (Project details at <https://doi.org/10.61686/FSWWF35053>).

IE&IS ESA Education Innovation Grant

EINDHOVEN UNIVERSITY OF TECHNOLOGY

2024

- Funding to develop an R package designed to assist master's students and supervisors in performing automatic checks to enforce best research practices.